

Write a Python program that takes a word as input. If its length is odd, print "Invalid". If even, split it into two equal halves and check if both halves have the same number of vowels. If they do, print "They have even vowel count", otherwise print "They do not have even vowel count"

```
def check_even_vowel_count(word):
    vowels = "AEIOUaeiou"

    if len(word) % 2 != 0:
        print("Invalid")
        return

    mid = len(word) // 2
    first_half, second_half = word[:mid], word[mid:]

    count_vowels = lambda s: sum(1 for char in s if char in vowels)

    if count_vowels(first_half) == count_vowels(second_half):
        print("They have even vowel count")
    else:
        print("They do not have even vowel count")

word = "helo"
check_even_vowel_count(word)
```

Write a Python program that takes n lines of input and capitalizes the first letter of each word in every line. Print the modified lines.

```
def capitalize_first_letter():
    n = int(input("Enter number of lines: "))
    lines = [input() for _ in range(n)]

    for line in lines:
        print(line.title())

# Example usage
capitalize_first_letter()
```

Count Words with a Given Prefix in a Sentence input : pay attend the
class at periodic time (sentence) at(prefix) output : 2"

```
def count_words_with_prefix(sentence, prefix):  
    words = sentence.split()  
    count = sum(1 for word in words if word.startswith(prefix))  
    return count
```

```
# Example usage  
sentence = input("Enter a sentence: ")  
prefix = input("Enter a prefix: ")  
print(count_words_with_prefix(sentence, prefix))
```

find the digit input 1:123hii output :no input2: 123456 output :yes

```
def contains_only_digits(s):  
    if s.isdigit():  
        return "yes"  
    else:  
        return "no"
```

```
# Test cases  
input1 = input()  
print("Output:", contains_only_digits(input1)) # Output: no
```

program that takes input from the user, calculates the sum of odd and even numbers from the input, and prints the result as a tuple

```
# Get input from the user as a list of numbers  
numbers = list(map(int, input("Enter numbers separated by space:  
").split()))
```

```
# Calculate the sum of odd and even numbers  
odd_sum = sum(num for num in numbers if num % 2 != 0)  
even_sum = sum(num for num in numbers if num % 2 == 0)
```

```
# Print the result as a tuple  
print(tuple((odd_sum, even_sum)))
```

program to find the longest word in a given string

```
def longest(sen):  
    words=sen.split()  
    longe=max(words,key=len)  
    return longe
```

```
sen=input()  
print(longest(sen))
```

print only even elements of a list, separated by spaces, without brackets

```
n=list(map(int,input().split()))  
  
print(*[num for num in n if num%2==0])
```

Python script that takes multiple sets of inputs, merges them into a single sorted list, and calculates the sum of the elements:

```
def merge():  
    n=int(input())  
    merges=[]  
  
    for i in range(n):  
        num=int(input())  
        elements=[int(input()) for t in range(num)]  
        merges.extend(elements)  
  
    merges.sort()  
    total=sum(merges)  
  
    print(merges)  
    print(total)  
  
merge()
```

Replacing the characters in the string with other characters or symbols

```
def replace_chars(text, replacements):

    for old_char, new_char in replacements.items():
        text = text.replace(old_char, new_char)
    return text

# Example usage:
original_text = "Hello, World!"
replacements = {'H': '#', 'e': '3', 'o': '0', 'l': '1', 'd': '$'}
new_text = replace_chars(original_text, replacements)
print(new_text)
```

Nth Fibonacci numbers

```
def fibonacci(n):
    if n <= 0:
        return "Invalid Input"
    elif n == 1:
        return 0
    elif n == 2:
        return 1
    else:
        return fibonacci(n - 1) + fibonacci(n - 2)

n = int(input("Enter a number: "))
print(f"The {n}th Fibonacci number is: {fibonacci(n)}")
```

Fibonacci numbers within range

```
def fibo(start, end):
    fibono=[]
    a,b=0,1
    while a<=end:
        if a>=start:
            fibono.append(a)
        a,b=b,a+b
    return fibono

a=int(input())
b=int(input())

print(fibo(a,b))
```

python code

input: 2 5(range)

output: [(2, 4), (3, 9), (4, 16), (5, 25)]

```
def gen_tuples(s,e):
    return [(i,i**2) for i in range(s,e+1)]

s=int(input())
e=int(input())
print(gen_tuples(s,e))
```

Remove consecutive repeated elements

Input: occurrence Output: ocurence

```
def dup(s):
    if not s:
        return "no dup"
    dupli=False
    result=[s[0]]
    for i in range(1,len(s)):
        if s[i]!=s[i-1]:
            result.append(s[i])
        else:
            dupli=True
```

```

        out="".join(result)
        return out if dupli else "NO dup"

s=input()
print(dup(s))

```

Count substring occurrences in a string

```

def count_substring_occurrences(string, substring):

    return string.count(substring)

# Example usage
main_string = "hello hello world hello"
sub_string = "hello"
count = count_substring_occurrences(main_string, sub_string)
print(f'The substring "{sub_string}" appears {count} times.')

```

check if a given string is a palindrome:

```

def is_palindrome(s):
    s = s.lower().replace(" ", "") # Convert to lowercase and
remove spaces
    return s == s[::-1]

# Example usage
word = input("Enter a word: ")
if is_palindrome(word):
    print("It's a palindrome!")
else:
    print("It's not a palindrome.")

```